

Introduction to Python

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What is Python Programming Language?

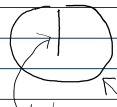
Python is a general purpose, dynamically typed, high level, compiled and interpreted garbage collected and purely object oriented and functional programming language.

We need to communicate with computing devices

Computing understands

binaries

two [0, 1]

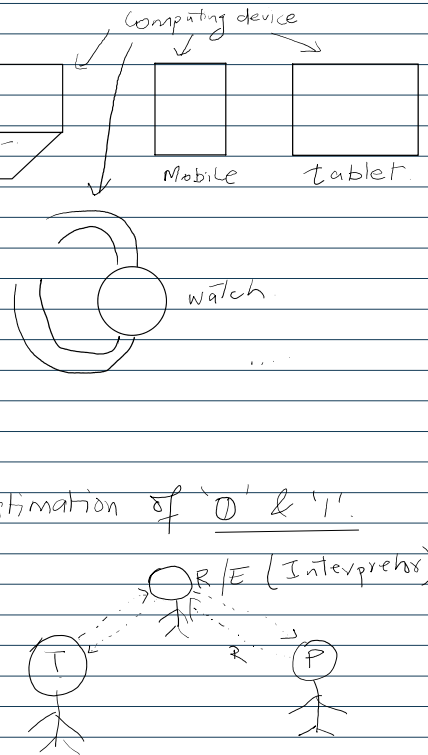


ground is '0' zero

off

Interpreter

We understand English. (Python)



→ print("hello, Python") ⇒

Interpreter

binary format

Generates the
Output

Process the
binary

Programmily
language
→ Dart
→ Go

What is a Program?

Program is a group of statements, they execute in a sequence.

- 1.
- 2.
- 3.

Program
{Python}

Python Compiler
/ Interpreter

Binary or
Bytecode

Output
in human
readable
format

Python Runtime
Environment
or
Python Virtual Machine
[PVM]

Features of Python and where Python is used.

- Easy to use and read.
- Dynamically typed.
- Then high-level.
- **Compiled and interpreted:** Python code first gets compiled into byte code and then interpreted line by line. When we download the python in our system, from python.org we download the default implementation of python known as Cpython. Cpython is considered to be compiled and interpreted both.
- Garbage collected.
- Purely Object Oriented.
- Cross-platform Compatibility.
- Open source.
- Extensible.
- Python has wide range of libraries.
- GUI programming support: It contains a library like Tinker and PyQt etc.
- Embeddable
- Large Community
- High Demand
- Big Data, Machine learning: NumPy, Pandas, SciKit-learn, tensorflow and many more.
- Development of Game: PyGame and many more.
- Computer vision or image processing.
- Speech recognition.
- Finance
- Audio and Music.

U I
↓ user Interface
Graphical

Python History and Version

- Python laid its foundation in late 1980s.
- The implementation of Python was started in December 1989 by Guido Van Rossum at CWI in Netherlands.
- In February, 1991 Guido Van Rossum published the code (labeled version 0.9.0) to alternate sources.
- In 1994 Python 1.0 was released with features like lambda, map, filter and reduce.
- Python 2.0 added in new features such as list comprehensions, garbage collection systems.
- Then on December 3, 2008 Python 3.0 (also called Pi3K.) was released. It was designed to rectify the fundamental flaw of the language.
- ABC programming language is set to be the predecessor of Python language, which was capable of exception handling and interface with the Amoeba (terminal) Operating System.
- The following programming language influenced python:
 - ABC language
 - Modula-3

came in 2008
↓
3.0.0
↓
3.13.3
↑
downloaded.
2025

→ Windows 11
→ Macintosh
→ ANDROID
→ IOS
→ LINUX
Thousands of OS are available

Why the name Python?

The name is based on show "Monte Python's Flying Circus". It was aired on BBC in 1970s. The show was liked by Guido. So he named the programming language Python.